Case Study: From Messy Data to Business Decisions

Excel Data Cleaning + Generative AI for Business Insights

Data Science Training Track — RetailNow Case Study

1. Business Scenario

You have just joined RetailNow, a mid-sized online retailer selling Electronics, Apparel, Home & Kitchen, Beauty, Sports, Groceries, Books, and Toys across India. The Operations Manager exports raw order data from the company's order system every quarter, but the export is notoriously messy — different regional teams enter data differently, and no one has cleaned it up.

The CEO wants a short business insights report before next week's leadership meeting. She has given you the raw export and asked you to use whatever tools you need — including AI — to get her clean numbers and clear recommendations fast.

2. The Dataset

File provided: RetailNow_Messy_Sales_Data.xlsx (415 order records, 14 columns). This is a single quarter of sales orders. It is intentionally messy — this mirrors what real analysts receive.

Column	Meaning	Known Issue(s)
Order ID	Unique order identifier	Some IDs repeat (duplicate orders)
Order Date	Date the order was placed	At least 5 different date formats, plus blanks
Customer Name	Customer full name	Mixed case, extra spaces, initials only, some blank
Email	Customer email	Some missing
City / Region	Delivery location	Inconsistent casing, typos, extra spaces
Product Category	Category of product ordered	Many spelling/casing variants of the same category
Product Name	Specific product ordered	Mostly clean
Quantity	Units ordered	Text values ("N/A", "two"), a suspicious 9999, negatives, blanks
Unit Price	Price per unit	Mixed currency symbols/text, thousands separators, blanks
Discount (%)	Discount applied	Mixed number/text/percent format, blanks
Payment Method	How the order was paid	Inconsistent casing and abbreviations
Order Status	Delivered / Shipped / Cancelled / Returned / Pending	Inconsistent casing
Customer Rating (1-5)	Post-purchase rating	Blanks and out-of-range values (0, 6, -1)

3. Your Task — Two Parts

Part A: Clean the Data in Excel

Using the Excel skills you've already learned (formulas, Text-to-Columns, Find & Replace, Power Query, data validation, conditional formatting to spot outliers), clean the dataset so it is ready for analysis. At minimum, you must:

1. Standardize Order Date into a single consistent date format.
2. Standardize Product Category, Region, Payment Method, and Order Status so each has exactly one spelling/casing per value (e.g., all "Electronics", not "electronics" / "ELECTRONICS" / "Electroncs").
3. Clean Customer Name (trim extra spaces, consistent capitalization).
4. Fix Quantity: handle text values, blanks, and the suspicious 9999 entries — decide and document a sensible rule (e.g., cap, flag, or remove).
5. Fix Unit Price: strip currency symbols/text and commas so it is a pure number.
6. Standardize Discount (%) into one consistent numeric format.
7. Identify and handle exact duplicate order rows.
8. Flag or handle out-of-range Customer Rating values (must be 1–5).

Keep your raw data untouched on one sheet, and do your cleaning on a copy — this is standard practice so your work is auditable.

Formula starting points — figure out where each one fits

- A function that removes leading/trailing spaces from text.
- A function that changes text to one consistent case (there's an UPPER, a LOWER, and one that capitalizes each word — pick what suits Product Category).
- A lookup or nested-IF approach to map messy category spellings to one clean value using a small reference table.
- A function to convert text-formatted dates into real Excel dates (hint: Power Query's 'Column From Examples' or 'Data > Text to Columns' can help too).
- A way to strip non-numeric characters (₹, $, commas) from Unit Price so it behaves like a number, not text.
- Conditional formatting or a helper column with a logical test to flag Quantity or Rating values outside a valid range.

Part B: Use Generative AI to Get to Insights Faster

Once your data is clean, use a GenAI tool (ChatGPT, Claude, or Copilot) as your analysis partner. You are not coding — you are prompting. Use it to:

- **Formula help:** Suggest specific Excel formulas or Power Query steps for a cleaning problem you're stuck on.
- **Insight generation:** Turn your cleaned summary tables (e.g., a pivot table of revenue by category and region) into a plain-language explanation of what's happening in the business.
- **Critical review:** Challenge your first draft of insights — ask the AI to point out what claims aren't well supported by the numbers.
- **Report writing:** Help you write a polished, board-ready executive summary from your bullet-point findings.

You must include, in an appendix, at least 4 of the actual prompts you used and a one-line note on what you changed or kept from the AI's response. This is to show your prompting process, not just the final output.

4. Building Your Analysis — Where to Look

Before you open a GenAI tool, build at least 2–3 pivot tables and 2–3 charts from your cleaned data. The AI is much more useful once you feed it real numbers instead of asking it to guess.

Pivot tables worth trying — decide which fields go where

- Total revenue broken down by Product Category and by Region — which combination is strongest and weakest?
- Order count and average order value by Payment Method.
- A count of Order Status values by Region — where are cancellations or returns concentrated?
- Average Customer Rating by Product Category.

Charts worth trying — match the chart type to the question

- A chart comparing revenue across categories — which type highlights the biggest vs. smallest bars clearly?
- A trend chart of orders or revenue over time (by month/week) — what shape of chart shows a trend?
- A chart showing the share of orders by Order Status — think about what shows 'parts of a whole' well.
- A chart comparing average rating across categories — is a small difference more visible as bars or as a line?

Angles to explore with the AI once you have your tables — don't just ask 'give me insights'

- Ask it to compare your two strongest and two weakest categories and suggest possible reasons, then check those reasons against your own data.
- Ask it what a sudden spike in cancellations or returns in one region might indicate, and what you'd need to check to confirm it.
- Ask it whether your discount and revenue numbers suggest discounting is working or just eating into margin.
- Ask it to draft one recommendation per insight, then push back and ask 'what could make this recommendation wrong?'

5. Deliverables

1. Cleaned workbook — a copy of the Excel file with a clean data sheet plus at least one pivot table/summary view (e.g., revenue by category, by region, by month).
2. Business Insights Report (1–2 pages) — written in business language for the CEO, covering: top-performing and under-performing categories/regions, patterns in cancellations/returns, any pricing or discount observations, and 3–5 concrete recommendations.
3. Prompt appendix — the GenAI prompts you used, as described in Part B.

6. Sample Starter Prompts (you should go beyond these)

- "I have a column of dates in Excel in at least 5 different formats, including some blanks. What's a reliable way to standardize them using Power Query?"
- "Here is a pivot table showing revenue by product category and region [paste/describe your numbers]. What are 3 business insights a retail CEO would care about?"
- "Review this insight: 'Electronics is our best category.' What data would I need to back this claim up, and what might I be missing?"
- "Rewrite these 5 bullet points as a short, professional executive summary for a CEO audience."